vsec> use vsec_new

Output

switched to db vsec_new

vsec_new> db.createCollection("students")

Output

{ ok: 1 }

vsec_new> show collections

Output

students

vsec_new> db.students.insertOne({

| name: "Preeti",

| age: 19,

| course: "IT"

| })

Output

{

acknowledged: true,

insertedId: ObjectId('6a867d08c99511d2a33682e1')

}

vsec_new> db.students.insertMany([

| { name: "Rahul", age: 21, course: "CS" },

| { name: "Ram", age: 23, course: "IT" },

| { name: "Sneha", age: 20, course: "IT" }

|])

Output

{

acknowledged: true,

insertedIds: {

'0': ObjectId('6a867d20c99511d2a33682e2'),

'1': ObjectId('6a867d20c99511d2a33682e3'),

'2': ObjectId('6a867d20c99511d2a33682e4')

```
  }
}
```

vsec_new> db.students.find()

Output

```
[
 {
   _id: ObjectId('6a867d08c99511d2a33682e1'),
   name: 'Preeti',
   age: 19,
   course: 'IT'
 },
 {
   _id: ObjectId('6a867d20c99511d2a33682e2'),
   name: 'Rahul',
   age: 21,
   course: 'CS'
 },
 {
   _id: ObjectId('6a867d20c99511d2a33682e3'),
   name: 'Ram',
   age: 23,
   course: 'IT'
 },
 {
   _id: ObjectId('6a867d20c99511d2a33682e4'),
   name: 'Sneha',
   age: 20,
   course: 'IT'
 }
]
```

vsec_new> db.students.find({

| name: "Preeti"

| })

Output

```
[
  {
    _id: ObjectId('6a867d08c99511d2a33682e1'),
    name: 'Preeti',
    age: 19,
    course: 'IT'
  }
]
```

vsec_new> db.students.find({

| course: "IT"

| })

Output

```
[
  {
    _id: ObjectId('6a867d08c99511d2a33682e1'),
    name: 'Preeti',
    age: 19,
    course: 'IT'
  },
  {
    _id: ObjectId('6a867d20c99511d2a33682e3'),
    name: 'Ram',
    age: 23,
    course: 'IT'
  },
  {
    _id: ObjectId('6a867d20c99511d2a33682e4'),
    name: 'Sneha',
```

```
    age: 20,
    course: 'IT'
  }
]
```

vsec_new> db.students.find({

| age: { $gt: 20 }

| })

Output

```
[
  {
    _id: ObjectId('6a867d20c99511d2a33682e2'),
    name: 'Rahul',
    age: 21,
    course: 'CS'
  },
  {
    _id: ObjectId('6a867d20c99511d2a33682e3'),
    name: 'Ram',
    age: 23,
    course: 'IT'
  }
]
```

vsec_new> db.students.find({

| age: { $lte: 20 }

| })

Output

```
[
  {
    _id: ObjectId('6a867d08c99511d2a33682e1'),
    name: 'Preeti',
    age: 19,
```

```
    course: 'IT'
  },
  {
    _id: ObjectId('6a867d20c99511d2a33682e4'),
    name: 'Sneha',
    age: 20,
    course: 'IT'
  }
]
vsec_new> db.students.updateOne(
|     { name: "Preeti" },
|     { $set: { age: 20 } }
| )
```

Output

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
vsec_new> db.students.updateOne(
|     { name: "Preeti" },
|     { $set: { marks: 92 } }
| )
```

Output

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
```

```
  upsertedCount: 0
}
```

vsec_new> db.students.updateMany(

| { course: "IT" },

| { $set: { department: "Information Technology" } }

|)

Output

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
```

vsec_new> db.students.find()

Output

```
[
  {
    _id: ObjectId('6a867d08c99511d2a33682e1'),
    name: 'Preeti',
    age: 20,
    course: 'IT',
    marks: 92,
    department: 'Information Technology'
  },
  {
    _id: ObjectId('6a867d20c99511d2a33682e2'),
    name: 'Rahul',
    age: 21,
    course: 'CS'
  },
```

```
  {
    _id: ObjectId('6a867d20c99511d2a33682e3'),
    name: 'Ram',
    age: 23,
    course: 'IT',
    department: 'Information Technology'
  },
  {
    _id: ObjectId('6a867d20c99511d2a33682e4'),
    name: 'Sneha',
    age: 20,
    course: 'IT',
    department: 'Information Technology'
  }
]
```

vsec_new> db.students.find().sort({

| age: 1

| })

Output

```
[
  {
    _id: ObjectId('6a867d08c99511d2a33682e1'),
    name: 'Preeti',
    age: 20,
    course: 'IT',
    marks: 92,
    department: 'Information Technology'
  },
  {
    _id: ObjectId('6a867d20c99511d2a33682e4'),
    name: 'Sneha',
```

```
  age: 20,
  course: 'IT',
  department: 'Information Technology'
 },
 {
  _id: ObjectId('6a867d20c99511d2a33682e2'),
  name: 'Rahul',
  age: 21,
  course: 'CS'
 },
 {
  _id: ObjectId('6a867d20c99511d2a33682e3'),
  name: 'Ram',
  age: 23,
  course: 'IT',
  department: 'Information Technology'
 }
]
```

```
vsec_new> db.students.find().sort({
|     age: -1
| })
```

Output

```
[
 {
  _id: ObjectId('6a867d20c99511d2a33682e3'),
  name: 'Ram',
  age: 23,
  course: 'IT',
  department: 'Information Technology'
 },
 {
```

```
  _id: ObjectId('6a867d20c99511d2a33682e2'),
  name: 'Rahul',
  age: 21,
  course: 'CS'
 },
 {
  _id: ObjectId('6a867d08c99511d2a33682e1'),
  name: 'Preeti',
  age: 20,
  course: 'IT',
  marks: 92,
  department: 'Information Technology'
 },
 {
  _id: ObjectId('6a867d20c99511d2a33682e4'),
  name: 'Sneha',
  age: 20,
  course: 'IT',
  department: 'Information Technology'
 }
]
```

vsec_new> db.students.find().limit(2)

Output

```
[
 {
  _id: ObjectId('6a867d08c99511d2a33682e1'),
  name: 'Preeti',
  age: 20,
  course: 'IT',
  marks: 92,
  department: 'Information Technology'
```

```
  },
  {
    _id: ObjectId('6a867d20c99511d2a33682e2'),
    name: 'Rahul',
    age: 21,
    course: 'CS'
  }
]
```

vsec_new> db.students.countDocuments()

Output

4

vsec_new> db.students.deleteOne({

| name: "Ram"

| })

Output

{ acknowledged: true, deletedCount: 1 }

vsec_new> db.students.find()

Output

```
[
  {
    _id: ObjectId('6a867d08c99511d2a33682e1'),
    name: 'Preeti',
    age: 20,
    course: 'IT',
    marks: 92,
    department: 'Information Technology'
  },
  {
    _id: ObjectId('6a867d20c99511d2a33682e2'),
    name: 'Rahul',
    age: 21,
```

```
  course: 'CS'
 },
 {
  _id: ObjectId('6a867d20c99511d2a33682e4'),
  name: 'Sneha',
  age: 20,
  course: 'IT',
  department: 'Information Technology'
 }
]
```

vsec_new> db.students.deleteMany({

| age: { $gt: 21 }

| })

Output

```
{ acknowledged: true, deletedCount: 0 }
```

vsec_new> db.students.findOne()

Output

```
{
 _id: ObjectId('6a867d08c99511d2a33682e1'),
 name: 'Preeti',
 age: 20,
 course: 'IT',
 marks: 92,
 department: 'Information Technology'
}
```

vsec_new> db.students.findOne({ name: "Preeti" })

Output

```
{
 _id: ObjectId('6a867d08c99511d2a33682e1'),
 name: 'Preeti',
 age: 20,
```

```
  course: 'IT',
  marks: 92,
  department: 'Information Technology'
}
```

```
vsec_new> db.students.find({
|    course: "IT",
|    age: 20
| })
```

Output

```
[
  {
    _id: ObjectId('6a867d08c99511d2a33682e1'),
    name: 'Preeti',
    age: 20,
    course: 'IT',
    marks: 92,
    department: 'Information Technology'
  },
  {
    _id: ObjectId('6a867d20c99511d2a33682e4'),
    name: 'Sneha',
    age: 20,
    course: 'IT',
    department: 'Information Technology'
  }
]
```

```
vsec_new> db.students.find({
|    $or: [
|       { age: 20 },
|       { age: 23 }
|    ]
```

| })

Output

```
[
 {
   _id: ObjectId('6a867d08c99511d2a33682e1'),
   name: 'Preeti',
   age: 20,
   course: 'IT',
   marks: 92,
   department: 'Information Technology'
 },
 {
   _id: ObjectId('6a867d20c99511d2a33682e4'),
   name: 'Sneha',
   age: 20,
   course: 'IT',
   department: 'Information Technology'
 }
]
```

vsec_new> db.students.find({

| course: { $in: ["IT", "CS"] }

| })

Output

```
[
 {
   _id: ObjectId('6a867d08c99511d2a33682e1'),
   name: 'Preeti',
   age: 20,
   course: 'IT',
   marks: 92,
   department: 'Information Technology'
```

```
  },
  {
    _id: ObjectId('6a867d20c99511d2a33682e2'),
    name: 'Rahul',
    age: 21,
    course: 'CS'
  },
  {
    _id: ObjectId('6a867d20c99511d2a33682e4'),
    name: 'Sneha',
    age: 20,
    course: 'IT',
    department: 'Information Technology'
  }
]
```

```
vsec_new> db.students.updateOne(
|     { name: "Preeti" },
|     { $inc: { marks: 5 } }
| )
```

Output

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
vsec_new> db.students.updateMany(
|     {},
|     { $unset: { department: "" } }
| )
```

Output

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 2,
  upsertedCount: 0
}
```

```
vsec_new> db.students.find({
|     name: { $regex: "^P" }
| })
```

Output

```
[
  {
    _id: ObjectId('6a867d08c99511d2a33682e1'),
    name: 'Preeti',
    age: 20,
    course: 'IT',
    marks: 97
  }
]
```

```
vsec_new> db.students.find()
|     .skip(1)
|     .limit(2)
```

Output

```
[
  {
    _id: ObjectId('6a867d20c99511d2a33682e2'),
    name: 'Rahul',
    age: 21,
    course: 'CS'
```

```
  },
  {
    _id: ObjectId('6a867d20c99511d2a33682e4'),
    name: 'Sneha',
    age: 20,
    course: 'IT'
  }
]
```

```
vsec_new> db.students.aggregate([
|   {
|     $group: {
|       _id: "$course",
|       totalStudents: { $sum: 1 }
|     }
|   }
| ])
```

Output

```
[
  { _id: 'IT', totalStudents: 2 },
  { _id: 'CS', totalStudents: 1 }
]
```

```
vsec_new> db.students.drop()
```

Output

```
true
```

```
vsec_new> db.students.insertMany([
|   { _id: 1, name: "Preeti", age: 20, course: "IT", marks: 92 },
|   { _id: 2, name: "Rahul", age: 21, course: "CS", marks: 35 },
|   { _id: 3, name: "Ram", age: 23, course: "IT", marks: 67 },
|   { _id: 4, name: "Sneha", age: 20, course: "IT", marks: 81 },
|   { _id: 5, name: "Amit", age: 19, course: "CS", marks: 30 }
| ])
```

Output

```
{
  acknowledged: true,
  insertedIds: { '0': 1, '1': 2, '2': 3, '3': 4, '4': 5 }
}
```

```
vsec_new> db.students.find({
|     age: { $gte: 21 }
| })
```

Output

```
[
  {
    _id: 2,
    name: 'Rahul',
    age: 21,
    course: 'CS',
    marks: 35
  },
  {
    _id: 3,
    name: 'Ram',
    age: 23,
    course: 'IT',
    marks: 67
  }
]
```

```
vsec_new> db.students.find({
|     marks: { $lt: 40 }
| })
```

Output

```
[
  {
```

```
  _id: 2,
  name: 'Rahul',
  age: 21,
  course: 'CS',
  marks: 35
 },
 {
  _id: 5,
  name: 'Amit',
  age: 19,
  course: 'CS',
  marks: 30
 }
]
```

```
vsec_new> db.students.find({
|    marks: { $gt: 40 }
| }).limit(2)
```

Output

```
[
 {
  _id: 1,
  name: 'Preeti',
  age: 20,
  course: 'IT',
  marks: 92
 },
 {
  _id: 3,
  name: 'Ram',
  age: 23,
  course: 'IT',
```

```
  marks: 67
 }
]
vsec_new> db.students.find().sort({
|    marks: 1
| })
```

Output

```
[
 {
  _id: 5,
  name: 'Amit',
  age: 19,
  course: 'CS',
  marks: 30
 },
 {
  _id: 2,
  name: 'Rahul',
  age: 21,
  course: 'CS',
  marks: 35
 },
 {
  _id: 3,
  name: 'Ram',
  age: 23,
  course: 'IT',
  marks: 67
 },
 {
  _id: 4,
```

```
  name: 'Sneha',
  age: 20,
  course: 'IT',
  marks: 81
 },
 {
  _id: 1,
  name: 'Preeti',
  age: 20,
  course: 'IT',
  marks: 92
 }
]
```

vsec_new> db.students.find().sort({ marks: -1 }).limit(3)

Output

```
[
 {
  _id: 1,
  name: 'Preeti',
  age: 20,
  course: 'IT',
  marks: 92
 },
 {
  _id: 4,
  name: 'Sneha',
  age: 20,
  course: 'IT',
  marks: 81
 },
 {
```

```
  _id: 3,
  name: 'Ram',
  age: 23,
  course: 'IT',
  marks: 67
 }
]
```

vsec_new> db.students.find().sort({ marks: -1 }).limit(1)

Output

```
[
 {
  _id: 1,
  name: 'Preeti',
  age: 20,
  course: 'IT',
  marks: 92
 }
]
```

vsec_new> db.students.find(

| {},

| { _id: 0, name: 1, marks: 1 }

|)

Output

```
[
 { name: 'Preeti', marks: 92 },
 { name: 'Rahul', marks: 35 },
 { name: 'Ram', marks: 67 },
 { name: 'Sneha', marks: 81 },
 { name: 'Amit', marks: 30 }
]
```

vsec_new> db.students.find(

```
|    {},
|    { _id: 0, name: 1, marks: 1 }
| ).sort({ marks: -1 }).limit(1)
```

Output

```
[
  { name: 'Preeti', marks: 92 }
]
vsec_new> db.students.updateOne(
|    { marks: { $lt: 40 } },
|    { $set: { marks: 40 } }
| )
```

Output

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
vsec_new> db.students.deleteOne({
|    marks: { $lt: 40 }
| })
```

Output

```
{ acknowledged: true, deletedCount: 1 }
vsec_new> db.students.deleteMany({
|    marks: { $lt: 40 }
| })
```

Output

```
{ acknowledged: true, deletedCount: 0 }
vsec_new>
```